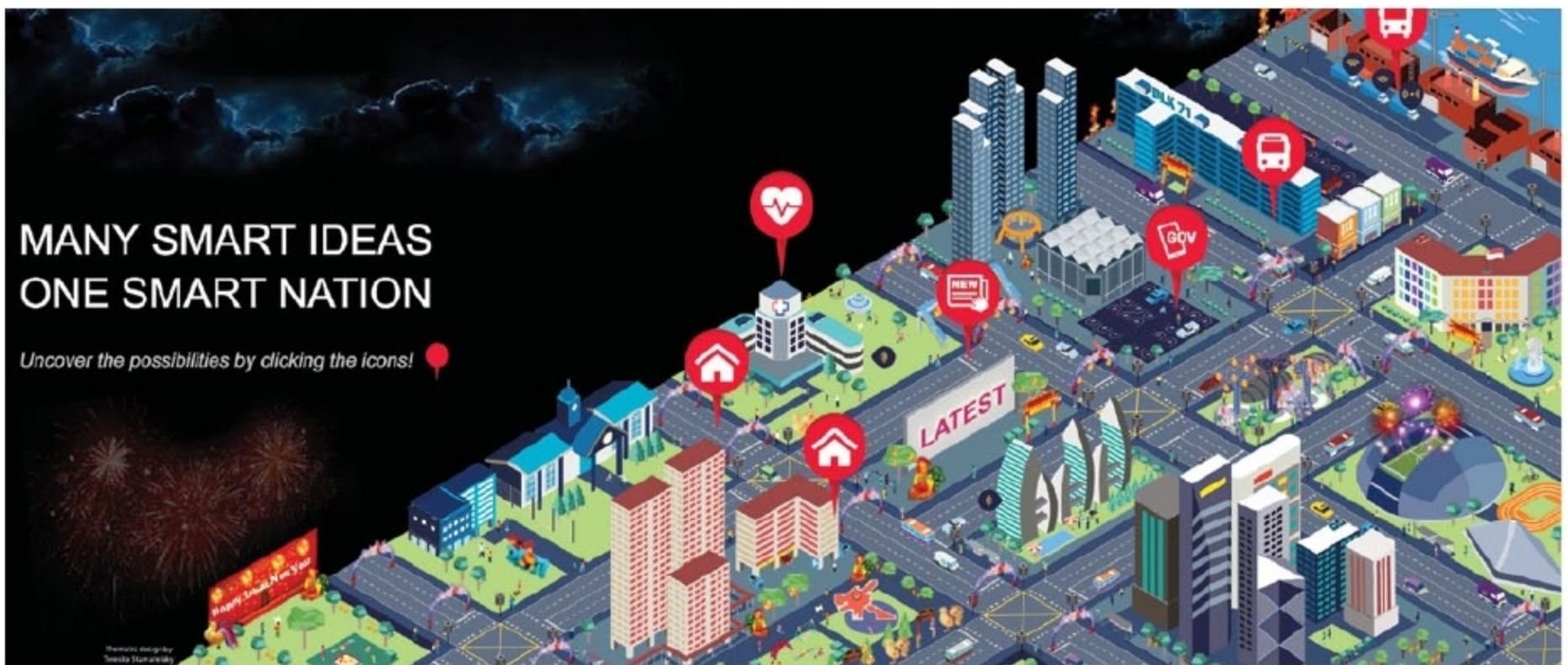




(Credit: <https://oss.adm.ntu.edu.sg>)

the world have adopted smart traffic solutions to control the flow of traffic. Road-surface sensors and CCTV cameras are installed to send real-time updates about traffic flow to a central traffic management system. Data feed from cameras are analysed and users are notified regarding congestion and traffic signal malfunctions.

Smart parking. Solving the traffic congestion issue is not enough; finding a parking slot especially during holiday time is a real struggle. Smart parking solutions can determine whether parking slots are free or occupied. Smart parking creates a real-time parking map by using road surface sensors embedded in the ground on parking area. This can reduce the time to wait, to find an empty slot for parking, which, in turn, helps in reducing congestion and pollution.

Smart waste management. Waste cannot be avoided but it can be properly managed. Waste management is an essential city service. With the ever-increasing urban population, there is a need to adopt smart methods for waste management. Use of AI technology not only provides smart waste management but also provides a sustainable waste management system. For instance, smart sensors fitted on waste bins send notifications to authorities to dispatch waste collection trucks as soon as they are about to be full.

Separate bins for paper, plastic, glass and waste food items should be maintained. These waste materi-

als can be recycled and used later for other useful purposes.

Smart policing. Nobody wants to live in a crime-ridden city. Smart policing ensures law enforcement agencies employ evidence-based and data-driven strategies to effectively and economically control in an efficient way. For example, a network of cameras and sensors can be installed in every corner to help identify people who commit crime, smoke in prohibited zones or seem suspicious. Cameras can be used to monitor crowd density, vehicle movement and cleanliness in a particular area, among others.

Smart governance. Smart city infrastructure is not complete without smart governance. Smart governance implies the use of information and communication technology intelligently to reduce corruption in governments and public administration organisations. The main goal is to improve decision-making through better collaboration among different stakeholders, including government and citizens. Smart governance uses data, evidence and other resources to improve decision-making and compliance for citizens.

Smart nation

Smart Nation concept was launched by the government of Singapore in 2014. Its main goal is to harness information and communication technologies, networks and Big Data tech to create tech-enabled solutions.

AI plays an important role in the journey towards creating a smart nation. Currently, Singapore has identified five national AI projects—transport and logistics, smart cities and estates, healthcare, education and safety and security. These projects can bring about many changes through the creation of new solutions to almost all the challenges faced today.

Digital technologies and AI are going to revolutionise the modern world in many ways including creating a smart society and smart city. Smart society, in modern context, is where people in a particular geographical area get all benefits and solutions through the use of technology, including easy transportation, protection, security, basic amenities and social well-being. A smart nation is ideally made up of smart societies and smart cities.

Conclusion

This article covered the importance of AI technology in smart homes and smart cities. Smart home refers to a modern home with all electrical appliances and devices controlled from a remote distance often via mobile apps and other smart devices that can operate interactively and autonomously. Smart cities employ technology in a way that positively impacts their inhabitants. Technology plays a key role in empowering smart citizens, strengthening smart societies, building smart cities and becoming smart nations. **EFY**